Alkanols

Methanol is the first in a series of structurally similar compounds called **alkanols** (alcohols). All alkanols have a name ending in '-ol'. They are similar in structure to alkanes, but they include a hydroxy (OH) group in the place of one hydrogen atom. Figure 1 shows the structural formula for methanol.

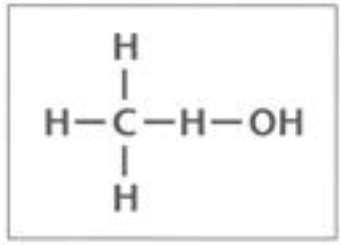


Figure 1

1 Give the molecular formula for methanol.

The next alkanols are ethanol, propanol and butanol. The structural formulae of propanol and butanol are shown in figure 2.

2 Ethanol has the formula CH_3CH_2OH. Draw the structural formula for ethanol.

3 Give the molecular formula for propanol.

4 Give the molecular formula for butanol.

5 The next alkanol in this homologous series is pentanol. Draw its structural formula and give its molecular formula.

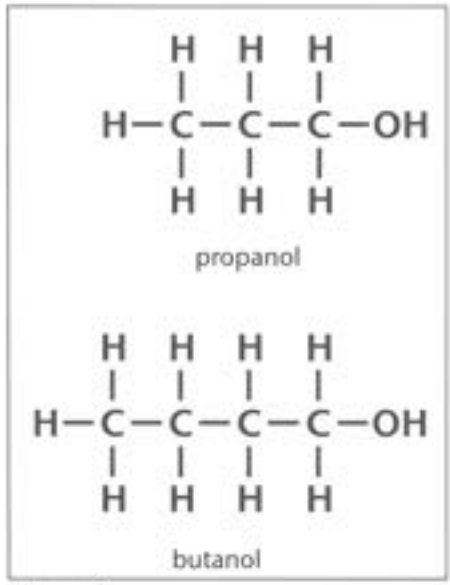


Figure 2

Common alkanols are liquid at room temperature and can be used as solvents, fuels and raw materials (with alkanoic acids) to make esters.

6 When ethanol is burned in excess oxygen, carbon dioxide and water are produced. Write a balanced formula equation for this reaction.

Alkanoic (carboxylic) acids

These organic acids are much weaker than common mineral acids such as sulfuric acid, nitric acid and hydrochloric acid. The first five alkanoic acids are methanoic, ethanoic, propanoic, butanoic and pentanoic acid. They are all liquids. Ethanoic acid is sometimes referred to as acetic acid. In dilute solution, it is vinegar. Figure 3 shows the structural formula for propanoic acid and butanoic acid.

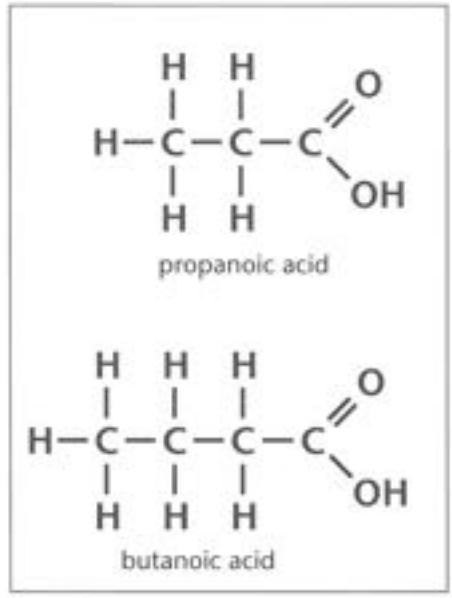


Figure 3

7 Draw the structural formula and give the molecular formulae for methanoic acid and ethanoic acid.

8 Give the molecular formula for propanoic acid and butanoic acid.

Figure 4 shows the functional groups of alkanols and alkanoic acids. Note that an **alkyl** is a group that is formed when a hydrogen atom is removed from an alkane.

9 Explain, with reference to alkanols and alkanoic acids, what is meant by a *functional group*.

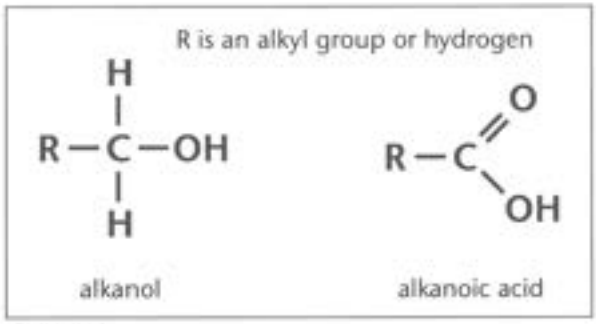


Figure 4

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